IAEA Coordinated Research Project on Development of Harmonized QA/QC Procedures for Maintenance and Repair of Nuclear Instruments

Test Procedure for Counter Systems and Timers for Nuclear Measurements

PROCEDURE Nº MRNI-513
REV. DO

Instituto Nacional de Investigaciones Nucleares
MÉXICO

DECEMBER 2008

Disclaimer:

The material in this document has been supplied by the authors and has not been edited by the IAEA. The views expressed remain the responsibility of the named authors and do not necessarily reflect those of the government(s) of the designating Member State(s). In particular, neither the IAEA nor any other organization or body sponsoring this meeting can be held responsible for any material reproduced in this document.

ELABORATED BY: FRANCISCO JAVIER RAMÍREZ JIMÉNEZ. **DATE**: DEC. 2008

REVIEWED BY: PEDRO CRUZ ESTRADA **DATE**: DEC. 2008

APPROVED BY: MARCO ANTONIO TORRES BRIBIESCA **DATE**: DEC. 2008

CONTENT

PAGE

1.- OBJECTIVE AND SCOPE 3

1.1.- Objective 3
1.2.- Scope 3

2.- NOTATION AND DEFINITIONS 3

2.1.- Notation 3
2.2.- Definitions 3

3.- INTRODUCTION 4

4.- DEVELOPMENT 4

4.1.- Generalities 4
4.2.- Test Instruments 4
4.3.- Test Conditions 5
4.4 .- Measurements 5

5.- ADMINISTRATION OF THE TEST REPORTS 12

5.1.- Numbering of the reports 12
5.2.- Personnel 12
5.3.- Test Report 12

6.- ACTION IN CASE OF NON CONFORMITIES 12

6.1.- Technical Report 12
6.2.- Labelling 12

7.- RESPONSIBILITIES 12

7.1.- Head of the Department 12
7.2.- Area Responsible 12
7.3.- Operative personnel 12

8.- BIBLIOGRAPHY 13

9.- ANNEXES 13

1.- OBJECTIVE AND SCOPE

1.1.- Objective

The objective of this procedure is to describe the test steps for the verification of performance and measurement of the characteristics of counter systems and timers used in nuclear applications in radiation detection chains of signals obtained from radiation detectors.

1.2.- Scope

These procedures are applicable to counter systems, scalers and timers used in Nuclear Instrumentation.

2.- NOTATION AND DEFINITIONS

2.1.- Notation

NIM	Nuclear Instrumentation Modules.
MCA	Multichannel Analyzer
LLD	Lower level discriminator
ULD	Upper level discriminator
DNL	Differential non linearity
INL	Integral non linearity
PHA	Pulse height analysis
MCS	Multi scaling mode analysis
E	Main voltage discrimination level related with the amplitude of a signal
ΔE	Window, voltage discrimination gap for signals to be considered in the analysis.
INT	Integral discriminator, integral mode
DIFF	Differential mode, SCA mode

2.2.- Definitions

2.2.1. Detector
A device that converts the energy of a photon or incident particle in an electric pulse.

2.2.2. Differential non linearity, DNL
Value that determines the maximum variation in channel width in a SCA.

2.2.3. Integral non linearity, INL
Value that determines the maximum deviation from a straight line in the relationship between discrimination level and applied voltage in a SCA.

2.2.4. Single Channel Analizer, SCA.
Nuclear instrument that provides a digital pulse every time the analog input pulse fulfills the established requirements of amplitude.

3. INTRODUCTION

The measurement of the height of the pulses produced by a radiation detector provides the information about the energy of the incident radiation, because in most of the cases, the pulse height is proportional to the energy delivered in the detector.
The quantity of radiation or radioactive material is proportional to the number of pulses generated in the radiation detectors, then the accurate measurement of pulses in nuclear counting systems is essential, it is needed to guaranty the exact counting of pulses and also the accuracy of the timing gate employed in the counting.
The nuclear counting systems can be operated in Pulse Height Analyzing (PHA) mode or in Multi-Scaling Counting (MSC) mode. In PHA mode we distinguish between single channel analyzer (SCA) systems and multi-channel analyzer (MCA) systems. Only MCA can be operated in PHA or MSC mode.
In general, a quality control should be made to ensure a proper operation of the counting systems, the main characteristics to be tested are:

- Count accuracy
- Clock accuracy
- Linearity against count rate
- Count-rate non-linearity
- Peak shift versus count rate
- Chi squared test

4.- DEVELOPMENT

The sequence of steps to verify the electrical characteristics of counter systems and timers is described in the next paragraphs. A flux diagram of the process is shown in Annex I.

4.1- Generalities

Whenever possible, refer to the test conditions recommended by the manufacturer in the sheet of specifications of the equipments, a technical report about the verification of the electrical characteristics of the counter systems and timers must be elaborated, including the circuit diagram, environmental conditions, testing set-up, and details of the test instruments.

4.1.1.- Verify that the counter system or timer is installed and connected according to the instructions of the manufacturer.

4.2.- Test Instruments

All the instruments employed in the tests must be calibrated and with a valid calibration certificate.

4.2.1 Pulse generator.

Produces voltage tail pulses with a precise relative amplitude that is selected with a calibrated dial and attenuation steps.

4.2.2 Spectroscopy amplifier.

Amplifies tail pulses and includes a semi-gaussian shaping in the pulse, use the unipolar output. A good long term stability is desired, i.e. better than 0.01 %/ºC.

4.2.3 Oscilloscope

Use an analog or digital oscilloscope.

4.2.4 Counter or Scaler

The number of counts per unit of time is measured with a counter or scaler for nuclear pulses, it generally includes a voltage discriminator to block low amplitude noise pulses. The input pulses to the counter or scaler must be positive.

4.2.5 Digital counter

A digital counter could assure the accuracy of the nuclear counter. p

4.3- Test Conditions

Generally the counter systems and timers need to reach its stable temperature, in order to have its best performance, a warming period of half an hour is enough.

4.3.1.- Temperature

A stable temperature is recommended to perform the tests, between 20º C and 25ºC.

4.4.- Measurements

Ideally, a reference set-up should be established in which the accuracy and precision is warranted, therefore, the test measurements should be made by comparison, as seen in the Fig.1. All counts coming from the generator should be registered in the system under test and in the reference set-up at the same time.

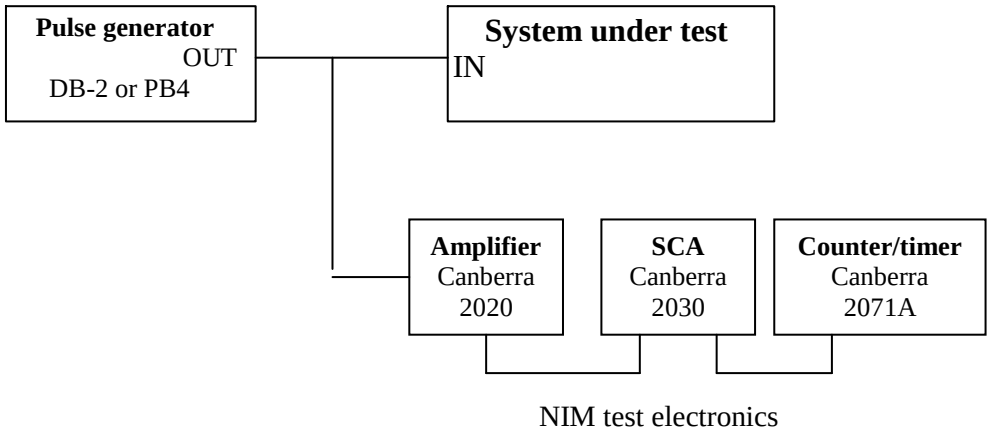


Fig. 1. Ideal set-up for the test of counting systems.

4.4.1 Count accuracy

The count accuracy of a system can be determined by using the set-up of Fig.1, follow the next steps:

- The reference set-up should have the same pulse shaping time as the system under test and should remain throughout all the tests.
- The system under test and the reference set-up are simultaneously started for counting, later the pulse generator has to be started manually and stopped before the selected counting time was reached. This procedure ensures that all pulses can be registered.
- Verify that the SCA settings for both system should permit a proper counting. When a system under test is operated in a MCS mode, all the contents of the various channels have to be summed-up. When the system operates in PHA mode, the integral spectra has to be taken.
- The maximum frequency of operation of the systems is limited by its processing time, therefore, it is good to make a set of measurements with different repetition rates of the pulses. Furthermore the system under test (MCA in PHA mode) has to be operated in real time mode. Record the results in the Table 1 of the Test Report.
- The deviation D in per cent is calculated as:

$$D = \frac{Cr - Ct}{Cr} \times 100 \qquad (1)$$

where: Cr = counts registered in the reference system
Ct = counts registered in the system under test

4.4.2. Clock accuracy

Clock accuracy is of great importance for a system applied in analytical measurements. The test set-up of Fig. 2 has to be used to identify the clock or timing deviation. Follow the next steps:

- Start the pulse generator
- All three systems (under test, NIM test electronics and reference digital counter) should be adjusted to the same counting time, and started almost simultaneously so that all systems see the same frequency jitter that could be included by the pulse generator.
- The pulses with a fixed repetition rate are registered in the system under test and in the NIM test electronic system consisting of amplifier, SCA and timer/counter. The deviation between the counter content of the NIM test electronics and the system under test is the clock error. A reference digital counter is used to determine any deviation in the time base of the NIM test electronics, which would mislead the results. One count deviation is not an

error, it is a result of the trigger mode in the counters or/and time jitter of the applied time base in the system under test. The Table 2 of the Test Report should present the results.

- The deviation D in per cent is calculated with equation 1.

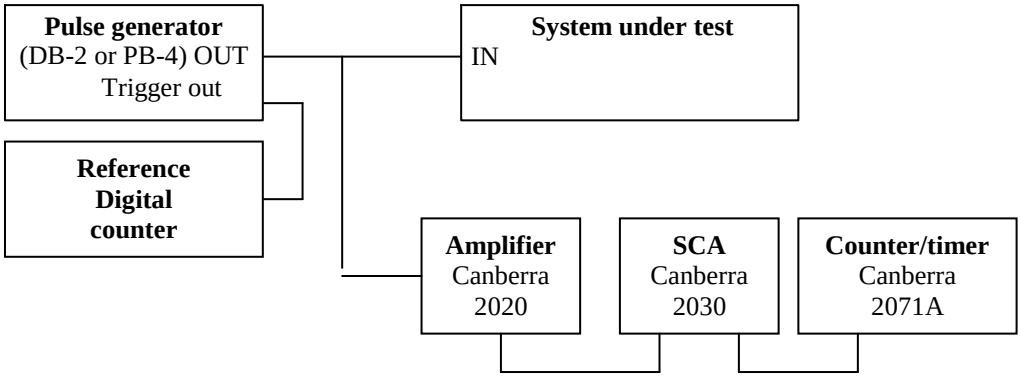


Fig. 2. Set-up for the test of clock accuracy.

4.4.3. Count-rate non-linearity:

The observation of count rate non-linearity, CRNL, is of high importance when the count-rate changes during an experiment so that the result of the total counting could include an error.

This test has to be performed using a random pulse generator, see Fig. 3. Both systems, NIM test electronic and system under test must have nearly the same shaping time to discard any error due to the difference of counting triggering. With higher count-rates, the probability of pile-up effects increases and some events would not be registered for example in a SCA system, because they would be are outside the SCA window setting. In MCA´s such events would be registered in a higher channel region. The reference counter/timer connected to the trigger out of the random pulse generator will measure the real number of counts and should verify when the NIM test electronic also loose counts, which will influence the interpretation of results.

- The count rate should be within +/- 5 %, it is not easy to adjust it in the generator. Obtain the best value of the count-rate using only the reference counter with a one second measurements and getting the average value for a set of at last 5 measurements. Table 3 collects the data to obtain the count-rate non-linearity. The CRNL in per cent is also calculated with equation 1.

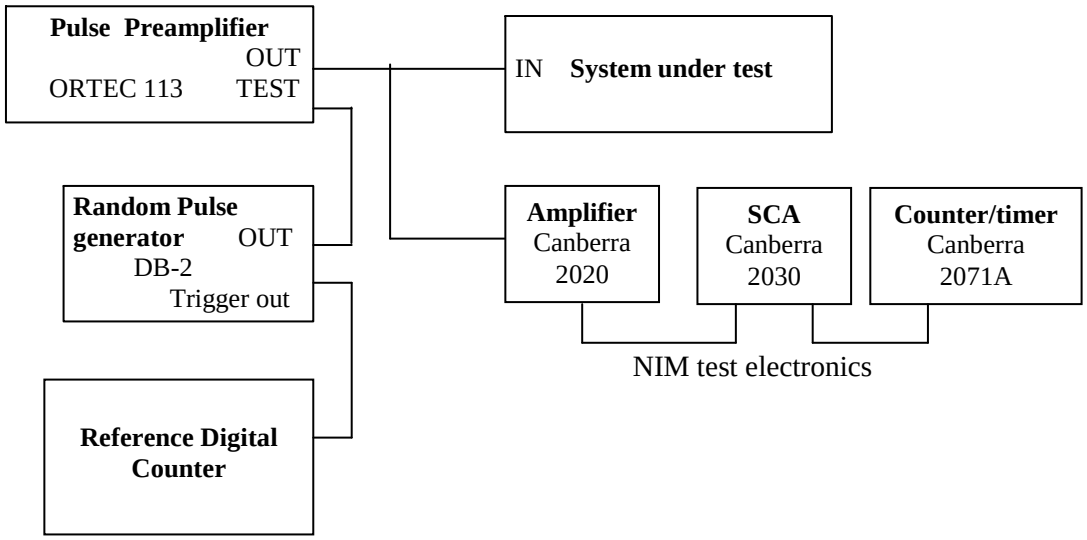


Fig. 3. Set-up for the test of count rate non linearity.

4.4.4. Peak shift versus count-rate.

Peak shifts in nuclear pulses due to high count-rates could affect the counts obtained in a counting system, the peak shifts could be caused by a poor base-line restoration in the spectroscopy amplifier. The evaluation the peak shift effects on the count rates obtained in a counting system can help to minimize errors in the measurements. The proposed set-up is shown in fig. 4.

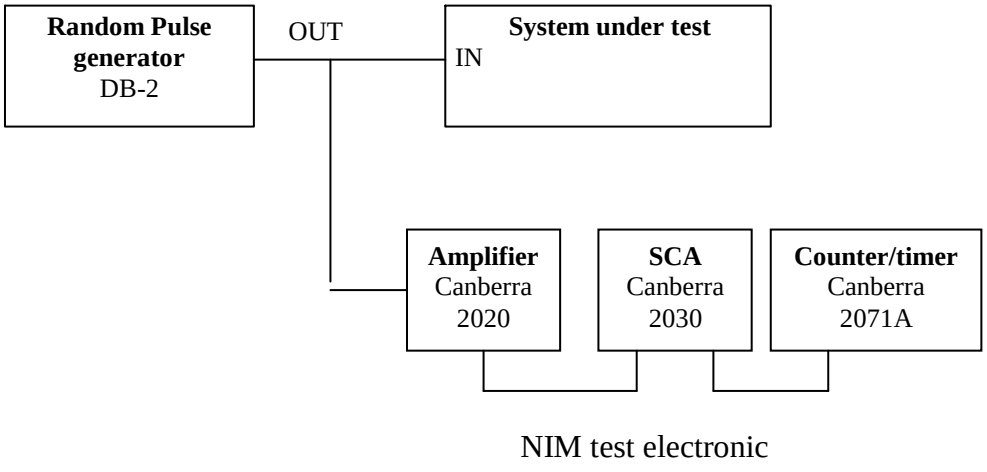


Fig. 4. Set-up for the measurement of the peak shift versus count rate.

The following steps are proposed for this measurement:

- The pulse generator is used in random mode and starts with a very low value of the amplitude Dial.
- The LLD of the two SCA´s is set to a high value, ≅ 5.0.
- For the low count rate value in Table 4 of the Technical Report, increase carefully the dial setting of the generator until the reference counter just starts counting, record this value of the dial, Do, in the Table 4, do the same procedure with the system under test.
- Repeat the former steps for each count-rate of the table, obtaining Di.
- Calculate the deviation, taking as reference the values of the generator dial obtained for the lower count rate:

$$D = \frac{Do - Di}{Do} \times 100 \qquad (2)$$

where: Do = Dial setting for the lower count rate.
Di = Dial settings for all the other count rates

4.4.5. Chi squared test:

This test is an overall Quality Control test and gives an indication of the proper operation of the counting system when processing random pulses from a radioactive source. For example, in a set of 10 measurements the Chi squared test results should be within 3.325 and 16.919 to consider that the measurements are within the normal statistical fluctuations, see the IAEA TECDOC 602.
When the result of the test is between the mentioned limits, this indicates that there is no instabilities in HV, SCA settings, amplifiers (base line shift or gain stability), counters (time base variations) and there is no influence by electromagnetic interference coming either from ground loops or interference with radio power stations or control signals for electrical devices. The Chi squared test depends only on the statistic pattern coming from a radioactive source.

In a series of N measurements X_i, when the mean value $\overline{X}$ is calculated as usual:

$$\overline{X} = \frac{\sum_{i=1}^{N} X_i}{N} \qquad (3)$$

the experimental variance is calculated with the following equation:

$$s^2 = \frac{1}{N} \sum_{i=1}^{N} \left(X_i - \overline{X} \right)^2 \qquad (4)$$

and the Chi squared is defined as:

$$\chi^2 = \frac{(N-1)\ s^2}{X} \qquad (5)$$

As an example of the application of this test utilizing a scintillation detector, the test set-up of Fig. is proposed .

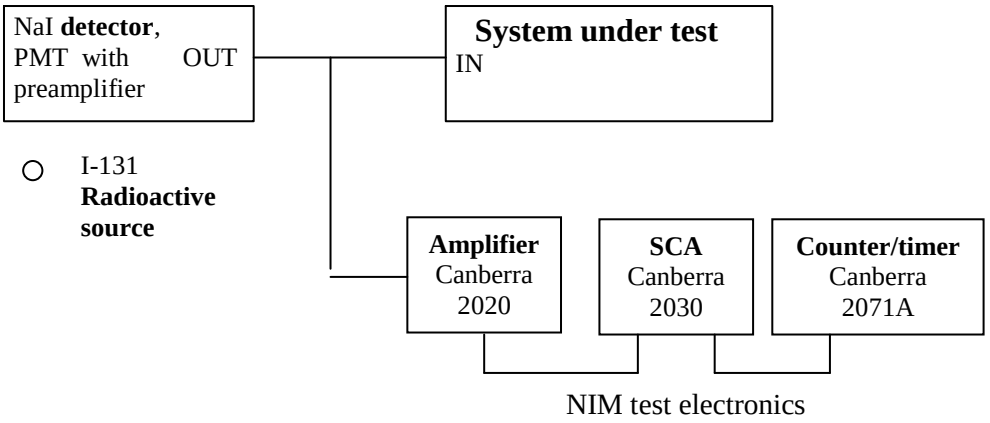


Fig. 5. Example of a set-up for the CHI square test.

The settings of the system under test and the NIM test electronic, shaping time, LLD and ULD have to be electronically the same. For purpose of demonstration in this example, a set of 10 measurements is considered. The counter of the NIM test electronic and the system under test have to be almost started simultaneously to avoid a deviation of the registered pulses, because they occur randomly in time. The 10 results of counting are shown in the Table A

The statistical analysis of all the values of Table A can be made with a reliable software, for example ORIGIN or EXCEL, that helps to calculate the Chi squared value. The obtained partial results can be stored in a file and database for purposes of traceability. The numbers included in Table A were calculated considering the experimental data of the *i* column.

i	X_i	$(X_i - \overline{X})$	$(X_i - \overline{X})^2$
1	**60692**	**-89.2**	**7956.64**
2	**60679**	**-102.2**	**10444.84**
3	**60593**	**-188.2**	**35419.24**
4	**60668**	**-113.2**	**12814.24**
5	**61076**	**294.8**	**86907.04**
6	**61063**	**281.8**	**79411.24**
7	**60959**	**177.8**	**31612.84**
8	**61033**	**251.8**	**63403.24**
9	**60547**	**-234.2**	**54849.64**
10	**60502**	**-279.2**	**77952.64**
	↓	↓	↓
Mean value $\overline{X}$	**60781.2**		↓
		$\sum_{i=1}^{N}(X_i - X)^2$	**460771.6**
		Chi squared χ^2	**7.5808243**

Table A. Collection of data for the calculation of the Chi squared value.

The Table 5 of the Test Report could be used to record the results of the Chi squared test of a counting system measured for different count rates, the average value of the count rate can be obtained as the mean value of a series of 10 measurements.

5.- ADMINISTRATION OF THE TEST REPORTS

5.1 Numbering of the reports.

All the generated technical test reports must have a unique and consecutive number.

5.2.- Personnel.

The test of detectors and electronic modules must be done by trained personnel.

5.3.- Technical Report of results.

5.3.1 The results of the test of detectors and electronic modules must be registered in a unique test report, stating the description of the detectors or/and electronic modules, mark, model, serial number, and all the test conditions, including the name of the person who made the tests.

5.3.2 All the test reports must be classified and keep in a folder for future consult.

6.- ACTION IN CASE OF NON CONFORMITIES.

6.1 Technical Report.

Even in the case that results of the test are not as expected, a technical report has to be elaborated, indicating the non conformities and how far are the measured characteristics from the ideal ones.

6.2 Labelling.

The components or equipments that are not under specifications or with a failure have to be marked with a label indicating: **OUT OF SPECIFICATIONS** and **FAILURE** respectively.

7.- RESPONSIBILITIES

7.1.- Head of the Department.

Supervise that all the activities for testing of detectors and electronic modules follow the established procedure.

7.2.- Area Responsible.

7.2.1 Assure that all the electronic test equipment be in good operational conditions and calibrated.

7.3.2 Verify that all the activities for testing of detectors and electronic modules follow the established procedure

7.2.3 Verify that the test reports contain all the details of the testing of detectors and electronic modules.

7.2.4 Maintain a register and control of the test reports for all the detectors and electronic modules tested in the laboratory.

7.3 Operative Personnel.

7.3.1 Verify that all the electronic test equipment be in good operational conditions and calibrated.

7.3.2 Follow the steps established in this procedure for the testing of detectors and electronic modules.

7.3.3 Elaborate the test report of all the tests of the detectors and electronic modules.

7.3.4 Inform to the Area Responsible of any anomalous condition encountered during the test procedure.

8.- BIBLIOGRAPHY

1.- ENGELS R., KAUFMANN H. "Control Test for Nuclear Counting Systems" Quality Control Procedures Applied to Nuclear Instruments, Proceedings of a Technical Meeting, IAEA HQ, Vienna, 23-24 August, 2007.

2.- ASTM "Standard Practice for Set-up, Calibration, and Quality Control of Instruments Used for Radioactivity Measurements" ASTM D7282-06 Standard, 2006.

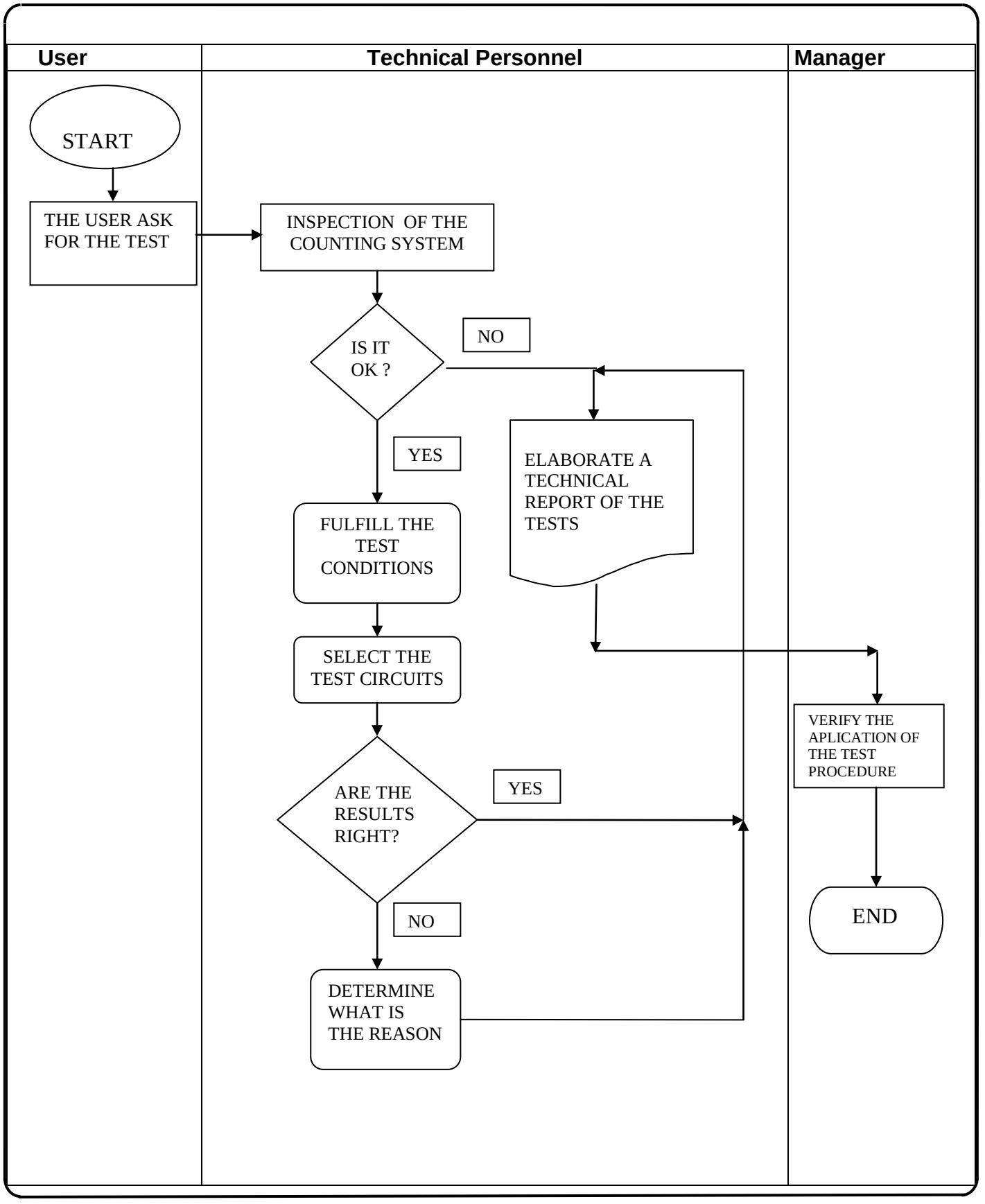
3.-. IAEA "Quality Control of Nuclear Medicine Instruments". TECDOC 602. IAEA. Viena, International Atomic Energy Agency, 1988.

4.- Knoll, Glenn F. "RADIATION DETECTION AND MEASUREMENT", Third Edition, John Wiley and Sons. U.S.A. 2000.

9.- ANNEXES

Annex I . Flow Chart
Annex II. Test Report

Annex I . Flow Chart


User
Technical Personnel
Manager
START
THE USER ASK FOR THE TEST
INSPECTION OF THE COUNTING SYSTEM
IS IT OK ?
NO
YES
ELABORATE A TECHNICAL REPORT OF THE TESTS
FULFILL THE TEST CONDITIONS
SELECT THE TEST CIRCUITS
ARE THE RESULTS RIGHT?
YES
NO
DETERMINE WHAT IS THE REASON
VERIFY THE APLICATION OF THE TEST PROCEDURE
END

Annex II. Test Report

TEST REPORT Nº CT_____

Counter system

(as required)

Instrument	Mark	Model	Serial number
Preamplifier			
Amplifier			
Counter -Timer			
SCA			
MCA			

Instruments Employed as reference

(as required)

Instrument	Mark	Model	Serial number
Preamplifier			
Amplifier			
Oscilloscope			
Pulse generator			
SCA			
Counter -Timer			
MCA			
Detector			
Digital Counter			

Environmental Conditions

Maximum change in temperature	Temperature

Test Circuits

Fig. 1	
Fig. 2	
Fig. 3	
Fig. 4	
Fig. 5	

Counting repetition rate	Counts of Reference System ***Cr***	Counts of System under test ***Ct***	Deviation D [%]
1 kHz			
2 kHz			
5 kHz			
10 kHz			
15 kHz			

Table 1: Count accuracy.

Repetition rate	Counts of reference counter	Counts of NIM test electronics ***Cr***	Counts of System under test ***Ct***	Clock deviation D [%]
1 kHz				
2 kHz				
5 kHz				
10 kHz				
15 kHz				

Table 2: Clock deviation

Count rate	Average value of Reference Counter (cps)	Counts of NIM test electronics **Cr**	Counts of System under test **Ct**	Count-rate non-linearity of the system under test **CRNL (%)**
100 cps				
200 cps				
500 cps				
750 cps				
1000 cps				
1500 cps				
2000 cps				

Table 3. Count-rate non-linearity.

Count rate	**Dial setting** of pulse generator when system under test starts counting	**Deviation D**	**Dial setting** of pulse generator when NIM test electronics starts counting	**Deviation D**
200 cps		0		0
500 cps				
1 kcps				
2 kcps				
10 kcps				

Table 4. Peak shift versus count rate

Count-rate	**System under test**		**NIM test electronics**	
	Average Count Number (C_{averge})	**Chi-square result**	Average Count Number (C_{averge})	**Chi-square result**
200 cps				
500 cps				
1000 cps				
2000 cps				

Table 5. Chi squared results.

Diagnostic or Comments:

Tested By:	
Date:	